

Recognise the connection between algebraic and graphical representations of exponential relations and solve related exponential equations, using digital tools where appropriate

Learning intention

- recognise the connection between algebraic and graphical representations of exponential relations and solve related exponential equations, using digital tools where appropriate.

Teach and model

- using digital tools with symbolic manipulation functionality to systematically explore exponential relations.
- recognising that in a table of values, if the ratio between consecutive values of the dependent variable is constant.
- investigating the links between algebraic and graphical representations of exponential functions using graphing software.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.